

Problem-Solution Fit

Date	02-July-2026
Team ID	xxxxxx
Project Name	Smart Lender
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] • Retail Loan Applicants • Credit Underwriters • Risk / Financial Analysts • Digital Banking Institutions	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] • Strict multi-variable form layout bounds • Inability to parse unmapped continuous feature skews • No historic data retention across page refreshes • Locked entirely to localized dev environment access structures	5. AVAILABLE SOLUTIONS PROS & CONS [AS] • Manual Verification / Traditional Rule Engines • Pros: Highly familiar, simple execution logic • Cons: Extremely slow, prone to script halts, no predictive scaling bounds
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] • Slow Screening: Manual validation creates bottlenecks (<i>Daily</i>) • High Opaque Friction: Applicant eligibility	9. PROBLEM ROOT / CAUSE [RC] • Decoupled, manual credit screening methods • Skewed historical datasets	7. BEHAVIOR + ITS INTENSITY [BE] • Applicants running eligibility checks before formal filing (<i>High</i>) • Underwriters reviewing diverse financial profile

uncertainty (<i>Per Application</i>) • Bias Risk: Class imbalances in historical bureau data (<i>Constant</i>) • Human Variance: Manual data entries introduce parsing errors (<i>Frequent</i>)	favoring majority profiles • Magnitude variations across continuous asset dimensions • Lack of lightweight web-integrated inference automation	attributes (<i>High</i>) • Users demanding ultra-low viewport response latency (<i>Critical</i>)
3. TRIGGERS TO ACT [TR] • Institutions demanding instant risk pre-screening • Surge in browser-based loan submission volumes • Need for automated, objective credit routing • Stricter metrics to reduce non-performing bank assets	10. YOUR SOLUTION [SL] Smart Lender: Local Flask app executing dynamic risk scoring. Features a SMOTE-balanced Random Forest classifier binary (<code>rdf.pkl</code>) and a standard feature pipeline scaler (<code>scale1.pkl</code>) to instantly process 11 exact inputs with sub-0.2 second latency and zero code context exceptions.	8. CHANNELS OF BEHAVIOR [CH] ONLINE: Localhost portal (<code>http://127.0.0.1:5000</code>), Git code repository
4. EMOTIONS BEFORE / AFTER [EM] • Before: Anxious, Overwhelmed, Uncertain, Stressed • After: Confident, Relieved, Satisfied, Secure		OFFLINE: Standard terminal workspace consoles, Local Anaconda environment loops